

5	(a)	(i)	$F = Bqv$ And Direction downwards (use FLHR)	
		(ii)	Magnetic force is perpendicular to the motion hence the kinetic energy of the particle is unchanged. Since K.E. is proportional to the square of the speed and mass, its speed and the magnetic force it experiences is constant. Since $F = Bqv$, the magnetic force is constant. Hence the path is circular.	
	(b)		The diagram in Fig. 5.2 shows a type of cathode ray tube containing a small quantity of gas. Electrons from a hot cathode emerge from a small hole in the conical shaped anode, and the path subsequently followed is made visible by the gas in the tube. The accelerating voltage is 5.0 kV.	
	(b)	(i)	Work done on electron = gain in K.E $qV = \frac{1}{2} m v^2$ $v = \sqrt{2qV / m}$ $= \sqrt{2 (1.6 \times 10^{-19}) (5000) / 9.11 \times 10^{-31}} = 4.19 \times 10^7 \text{ m s}^{-1}$	
		(ii)1	The magnetic force provides the centripetal force $Bqv = mv^2/r$ $r = mv / Bq$ $= 9.11 \times 10^{-31} \times 4.19 \times 10^7 / 2.0 \times 10^{-3} \times 1.6 \times 10^{-19}$ $= 0.12 \text{ m}$	
		2.	Condensation of the vapour/alcohol in the tube occurs and this forms a visible track along the trails of the charged particles (β -particles or α -particles or gamma ray photons). When the charged particles collide electrostatically with the gas atoms, they are ionized and the ions cause condensation of the vapour.	

6	(a)	(i)	$^{14}_7\text{N} + ^4_2\text{He} \rightarrow ^{17}_8\text{O} + ^1_1\text{H}$	
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